MILITARY
SPECIFICATION

MIL-G-12368(Sig C)
13 October 1952
Supersedes 71-706B
14 February 1944

GENERATORS GN-41 AND GN-41-B

1. SCOPE

1.1 This specification covers two types of hand operated ringing generators designated Generator GN-41 and Generator GN-41-B.

2. APPLICABLE SPECIFICATIONS, STANDARDS, DRAWINGS, AND PUBLICATIONS

2.1 The following specifications, standards and drawings of the issue in effect on the date of invitation for bids, form a part of this specification:

SPECIFICATION

FEDERAL

QQ-S-571	Solder; Soft (Tin, Tin-Lead and Lead-Silver)
UU-T-111	Tape, Paper, Gummed (Sealing and Securing)

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JAN-P-105	Packaging and Packing for Overseas Shipment - Boxes, Wood, Cleated, Plywood
JAN-P-106	Packaging and Packing for Overseas Shipment - Boxes, Wood, Nailed
MIL-B-107	Boxes, Wood, Wirebound (Overseas Type)
JAN-P-108	Packaging and Packing for Overseas Shipment - Boxes, Fiberboard (V-Board and W-Board), Exterior and Interior
MIL-P-116	Preservation, Methods of
JAN-P-127	Packaging and Packing for Overseas Shipment - Tape, Adhesive, Pressure-Sensitive, Water-Resistant
MIL-B-131	Barrier-Material, Water-Vaporproof, Flexible
JAN-T-152	Treatment, Moisture and Fungus-Resistant, of Communications, Electronic and Associated Electrical Equipment; General Process For
MIL-T-10513	Tropicalization of Materials Used in Signal Corps Equipment

STANDARDS

MILITARY

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	Marking of Shipments

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DRAWINGS

SIGNAL CORPS

460-1	Control Test Sampling Plans For 5% Defective Acceptable Quality Level
460-2	Control Test Sampling Plans For 5% Defective Acceptable Quality Level
SC-D-15914	Moisture-Resistance Test Cycle For Ground Signal Equipment, 48 Hour.
SC-DL-67371	Generators GN-38, GN-41, GN-38-B, GN-41-B

(Copies of specifications, standards and drawings required by contractors in connection with specific procurement functions should be obtained from or as directed by the contracting officer. Both the title and identifying number or symbol should be stipulated when requesting copies.)

2.2 Other publications.- The following publications, of the issue in effect on date of invitation for bids, unless otherwise stated, form a part of this specification:

American Standards Association

C8.7	Enameled Round Copper Magnet Wire

(Application for copies should be addressed to The American Standards Association, Inc., 70 East Forty-Fifth Street, New York 17, New York.)

3. REQUIREMENTS

3.1 Construction.- All construction shall conform to Drawing SC-DL-67371.

3.1.1 Coils.- Coils shall be evenly wound on the armature core.

3.1.1.1 Wire.- The wire used shall be enameled round copper magnet wire in accordance with ASA Standard C8.7, or Formex wire as made by General Electric Co., or equal. The size of wire and number of turns shall be such that the output requirements of 3.4 are met, but in no case shall the wire be smaller than 39.8 circular mils.

3.1.1.2 Insulation on core.- Before winding the coil, the armature core shall be well insulated with varnished cambric secured to the core by baked varnish; scotch acetate cloth electrical tape, as made by the Minnesota Mining and Manufacturing Co., or equal, may be used, provided an affidavit is furnished guaranteeing that the tape will not affect the enamel on the wire.

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3.1.1.3 Insulation between layers.- Two sheets of glassine paper, each 0.001 inch thick, shall be provided between layers of the winding at approximately 1000-turn intervals, but may be omitted if Formex wire, or equal, is used.

3.1.1.4 Inner lead.- The inner lead of the winding shall be insulated from the turns with impregnated cotton sleeving.

3.1.1.5 Impregnation and coating.- The windings shall be vacuum dried and impregnated with an insulating compound. The compound used shall not cold-flow at 160° F. nor be brittle at -80° F. The compound and the application temperature shall not injure the enamel on the wire at the time of impregnation or subsequently. After impregnation, the coil shall be covered by a soft cotton wrap and protected by a thorough coating of cellulose acetate lacquer.

3.1.1.6 Coil insulation.- Impregnated and coated coils shall meet the insulation tests of 4.3 before assembly in the armature.

3.1.1.7 Connections.- The inner lead of the winding shall be soldered to the grounded terminal at the pinion end of the armature, and the oute. lead shall be soldered to the insulated terminal at the other end.

3.2 Insulation material.- Insulation material which will support rapid combustion shall not be used. The use of inflammable varnish is prohibited. All insulating material except wire insulation shall be mechanically rigid at any temperature likely to be encountered in service and shall be free from cold-flow effects. Insualtion shall withstand the dielectric test of 4.8.

3.3 Magnet.- The magnets shall be properly aged in accordance with standard commercial practice for the particular type of magnet material used.

3.4 Output.- The output of each generator at a rotor speed of 1000 r.p.m. (which corresponds to approximately 212 r.p.m. of the crank) shall be not less than the values specified below for the stated resistance load.

Resistance Load, Including Resistance of the Milliameter OHMS	Output (Milliamperes)
200 ±1	140.0
100,000 ±100	0.87

3.5 Solder, flux and soldering.- Materials and process for soft soldering shall conform to the following:

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3.5.1 Soft solder.- Soft solder used for electrical connections shall be composition SN60 conforming to Specification QQ-S-571.

3.5.2 Flux and cleaning agents.- Flux for soldering shall be rosin or rosin and alcohol. No acid or acid salts shall be used in preparation for or during soldering; however, exception is permitted for preliminary tinning of electrical connections but in no case shall acid or acid salts be used where they can come in contact with insulation material. Where acid or acid salts are used, as permitted above, they shall be completely neutralized and removed immediately after use.

3.5.3 Soldering.- Soldered connections shall be neat. There shall be no sharp points or rough surfaces resulting from insufficient heating. The solder shall feather out to a thin edge, indicating proper flowing and wetting action, and shall not be crystallized, overheated, or underheated. The minimum necessary amount of flux and solder shall be used for electrical connections. Wherever practicable, excess rosin shall be removed with a wire brush and then a dry cloth; any resulting loose flakes of rosin shall be carefully removed from the inside of the equipment. Insulation material that has been subjected to heating during the soldering operation shall be undamaged and parts fastened thereto shall not have become loosened.

3.5.4 Cleaning.- Metal parts, after fabrication shall be cleaned in accordance with good commercial practice. Cleaning process shall have no visible or latent deleterious effect on the equipment. Corrosive material shall be removed completely before parts are assembled into the equipment. After assembly, components shall be cleaned thoroughly and shall be free from particles of solder, flux, and other foreign material. If necessary, such cleaning shall also be performed before final assembly of components.

3.6 Tropicalization.- The assembled equipment shall be treated in accordance with Specification JAN-T-152. Before being assembled into the equipment, materials and parts shall be treated in accordance with MIL-T-10513.

3.7 Service conditions.- The generator shall be capable of passing the service conditions tests of Section 4.

3.8 Workmanship.- Subassemblies, and parts shall be manufactured and assembled in a thoroughly workmanlike manner.

3.9 Preproduction samples.-

3.9.1 Quantities of preproduction samples.- After award of contract, these preproduction samples of the generator on order shall be furnished for performance testing. (See 6.2.(e).)

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3.9.2 Compliance with specified requirements.-

3.9.2.1 Preproduction samples shall meet specified requirements, shall be made and assembled by tools and methods that will be used for quantity production, and shall be accompanied by a statement to that effect. (See 3.9.2.3 for exception.)

3.9.2.2 Approval of preproduction samples shall not be construed as a waiver of any specified requirement. After being released to the contractor (see 3.9.4), preproduction samples to be offered as units on contract shall be refabricated by the contractor if necessary to meet specified requirements.

3.9.2.3 When deviation from 3.9.2.1 is unavoidable, the preproduction samples may be submitted for approval provided that the accompanying statement describes in detail each nonconforming feature, reason therefor, and manner in which it will be corrected in production of equipment on contract.

3.9.3 Test data.- The preproduction samples shall be accompanied by test data showing compliance with specified performance. The test data shall comprise an engineering report giving test procedure, observations, and other data, calculations, test results, and essential details of the testing equipment (manufacturer's model, serial number, date of calibration, and the like).

3.9.4 Reference standards.- After preproduction samples have been approved and returned to the contractor's plant, they shall be kept intact in custody of the Government inspector until released by him. They shall be used as reference standards to resolve any differences of opinion regarding interpretation of specified requirements.

4. SAMPLING, INSPECTION AND TEST PROCEDURES

4.1 Inspection, general.- Inspection shall consist of group A and B inspection as listed in Table I and as specified in 4.1.1 and 4.1.2. (See 6.3.1.) Group B inspection shall be performed on lots that have passed group A inspection and on samples selected from units that have been subjected to and met the group A inspection. However, this order of inspection may be varied when the Government inspector consideres it more pratical for administrative reasons to select samples for group B inspection. Samples for inspection shall be selected at random by the Government inspector.

4.1.1 Group A inspection.- At the discretion of the Government inspector, group A inspection shall be conducted by either or both of the following methods:

(a) 100 percent inspection.

(b) Statistical sampling in accordance with Military Standard MIL-STD-105. AQL's (acceptable quality levels) shall be as shown in table I or the contract.

Order of inspection: Any order which is satisfactory to the Government inspector may be used.

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4.1.2 Group B inspection.- The AQL's for group B inspection shall be as shown in Table I or the contract. Sampling plans for specified AQL's shall conform to Drawings 460-1 and 460-2.

4.1.2.1 Rejection.- If the number of defective sample units from a lot exceeds the specified acceptance number, the lot shall be rejected and the contractor shall take immediate action to correct the cause of failure.

4.1.2.2 Order of inspection.- Group B inspection shall be performed in the following order:

a. Overload

b. Peak voltage

c. Dielectric

4.1.3 Disposition of rejected product.- If a defective unit or lot is resubmitted for inspection, the unit or lot shall be suitably tagged or identified by equivalent means to indicate the cause of failure and means employed to correct the fault. The record shall be presented to the Government inspector when the unit or lot is resubmitted and shall become the property of the Government.

4.1.3.1 Groups A and B.- Disposition of product rejected during group A, or group B inspection shall be in accordance with Standard MIL-STD-105.

TABLE I - Inspection Procedures

Inspection	Requirement Paragraph	Test Paragraph	Preproduction	Group A AQL	Group B AQL
Visual and Mechanical, Major	---	4.2	X	1%	---
Visual and Mechanical, Minor	---	4.2	X	4%	---
Coil Insulation	3.1.1.6	4.3	X	1%	---
Output	3.4	4.4	X	1%	---
Overvoltage	---	4.5	X	4%	---
Overload	---	4.6	X	---	5%
Peak Voltage	---	4.7	X	---	5%
Dielectric	3.2	4.8	X	---	5%
Shake Test	---	4.9	X	4%	---
Service Conditions Tests	3.7	4.10	X	---	---

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4.2 Visual and mechanical inspection.- Parts and equipment shall be inspected for workmanship, mechanical fit, loose nuts and screws, application of specified tropicalization treatments, and miscellaneous defects. Controls and fastening devices shall be inspected for mechanical operation. Wiring, soldered connections, ground connections, welds, finishes, etc. shall be inspected for workmanship. Clearances, dimensions, and mechanical adjustments shall be measured. Labels and markings shall be inspected for conformance with approved facsimiles, when facsimilies are required. Visual and mechanical defects shall be classified by the Government inspector as major or minor in accordance with the definitions of Standard MIL-STD-105.

4.3 Coil insulation test.- Coil insulation shall be tested to withstand 330 volts r.m.s. a.c. between winding and core for 30 seconds. Coils shall also be tested for short-circuited turns.

4.4 Output.- The generator shall be tested for compliance with 3.4.

4.5 Overvoltage test.- The generator shall be driven for 15 seconds at a crank speed of 424 revolutions per minute at no load, without causing breakdown of insulation or damage to any part.

4.6 Overload and electrical demagnetization tests.- The generator shall be driven for 15 seconds at an armature speed of 2000 rpm while short circuited, without causing breakdown of insulation, damage to any part, or reduction in output of more than 5 percent below the values specified in paragraph 3.4. The magnets shall be remagnetized after this test.

4.7 Peak voltage test.- The generator shall be tested for peak voltage in the circuit shown in Figure 1. The instantaneous peak voltage at a rotor speed of 1000 rpm (crank speed 212 rpm), shall not exceed 270 volts. The peak voltage is equal to five times the microampere reading of the meter in the voltmeter circuit.

4.8 Dielectric test.- The dielectric strength of all circuits shall be tested to withstand 300 volts, or two times the peak operating voltage, whichever is the greater. The test voltage shall be d.c. or peak a.c. of the same value and shall be applied for 10 seconds.

4.9 Shake test.- The equipment shall be placed in its normal operating position on the table of the Package Tester made by the L.A.B. Corp., Summit, New Jersey, or equal. The package tester, shafts in phase, shall have its speed increased until it is possible to insert a 1/32 inch thick strip under one corner or edge of the equipment to a distance of 3 inches as it vibrates. The equipment shall be subjected to this test for 1 minute. The equipment shall meet full specification requirements after this test, and show no physical damage.

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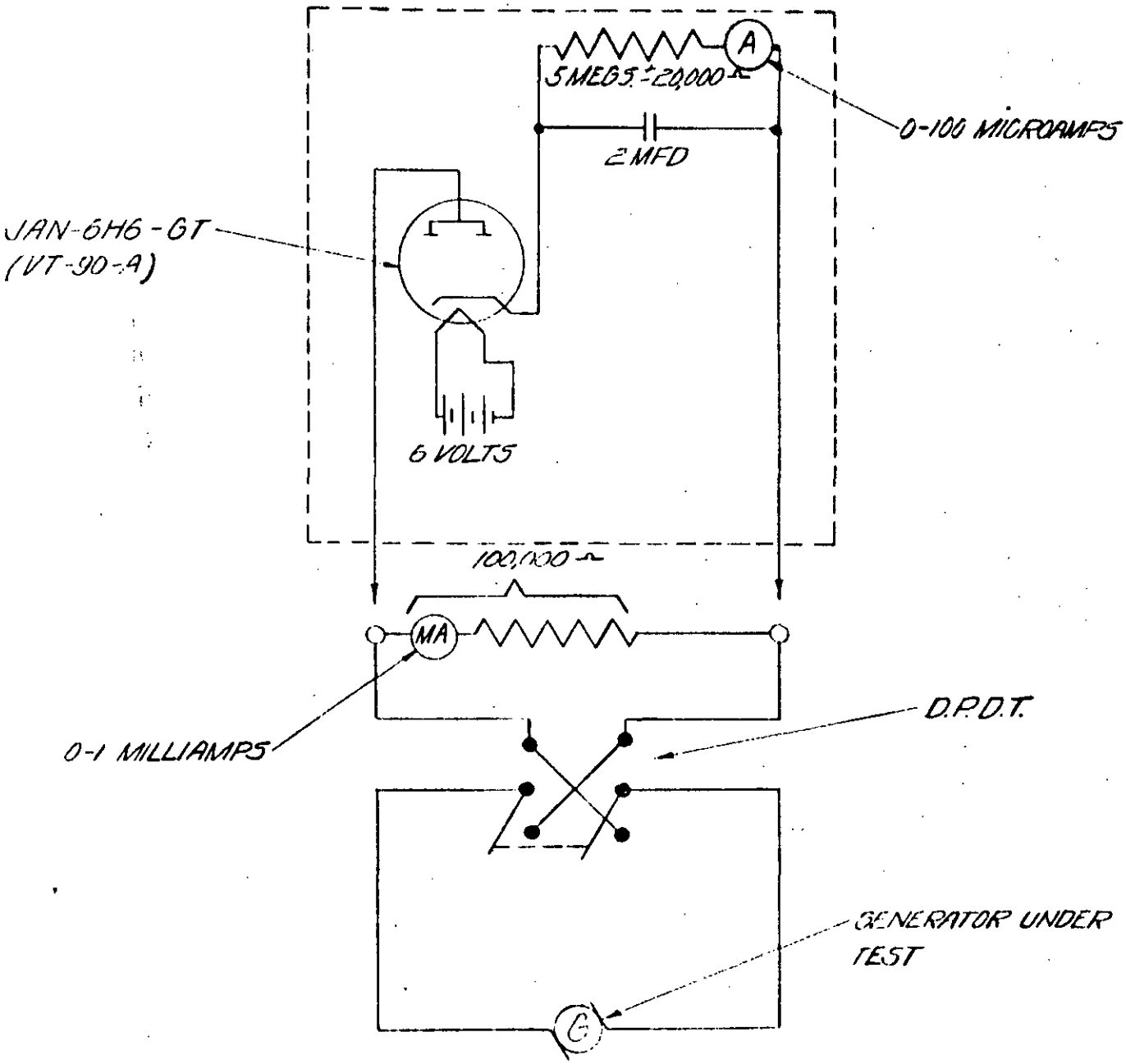


FIGURE I

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4.10 Service conditions tests.-

4.10.1 Low temperature test.- The equipment shall be maintained for 3 days in a chamber at an ambient temperature of -80° F. The temperature shall then be raised to -40° F. and stabilized for one hour. It shall then be subjected to the appropriate tests for performance measurement and shall meet full specification requirements.

4.10.2 High temperature test.- The equipment shall be maintained for three days in a chamber at an ambient temperature of + 160° F. The temperature shall then be lowered to + 131° F. and stabilized for one hour. It shall then be subjected to the appropriate electrical tests for performance measurement and shall meet full specification requirements.

4.10.3 Bounce test.- The generator, as packaged for shipment shall be placed on the table of the Package Tester as made by L.A.B. Corp., Summit, New Jersey, or equal, and shall be constrained from horizontal motion of more than 2 inches by suitable wooden fences. The package tester, shafts in phase, shall be operated at a speed of 285 rpm ± 1% for a total of 3 hours. During each 1/2 hour period of the test, the packaged equipment shall rest on a different face. The equipment shall meet full specification requirements after this test. Superficial surface abrasion is permissible.

4.10.4 Drop test.- The generator shall be dropped a distance of 4 feet on each face on four different positions of the parimeter. Drops may be made from a quick release hook, by an Acme Drop Tester, by a 10 degree Inclined Plane (Conbur) Tester, or by a Basket Type Pendulum Tester. The barrier or floor receiving the impact shall be 2 inch fir backed by concrete or by a rigid steel frame. After this test the equipment shall meet full specification performance requirements with no physical damage except surface abrasion.

4.10.5 Moisture-resistance.-

4.10.5.1 Test procedure.- The equipment shall be tested as follows:

a. Dry at + 130° F. ± 5° F. for 24 hours.

b. Condition at + 77° F. to ±5° F. and 40 to 50 percent relative humidity for 24 hours.

c. Take measurements as specified in 4.4 and readjust or realign as necessary to meet full specification requirements.

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4.10.5.1 Test procedure (Contd)

d. Subject to continuous cycling for five 48-hour cycles. Temperature, relative humidity, and period of time for each portion of the cycle shall conform to Drawing No. SC-D-15914. The test specified in 4.4 shall be made during the periods shown on the drawing and the output shall comply with 3.4.

e. After cycling has been completed, condition the equipment for 24 hours at + 77° F. ± 5° F. and 40 to 50 percent relative humidity. Then adjust for optimum performance, using only those means provided by the equipment. No repair or replacement of parts shall be made. After adjustment, the equipment shall meet full requirements for the test specified in 4.4.

4.10.5.2 Failure.- If the equipment fails to meet the performance specified in 4.4 during cycling, it shall be realigned or readjusted once. If the equipment then fails to meet 4.4 or fails subsequently during cycling, it does not pass the test.

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5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging.-

5.1.1 For domestic shipment.- Generators GN-41 and GN-41-B shall be preserved and packaged in accordance with Method III of Specification MIL-P-116.

5.1.2 For export shipment.- Preserve and package Generators GN-41 and GN-41-B in accordance with procedures specified for the designated methods as prescribed in Specification MIL-P-116, and further described in the following paragraphs.

5.1.2.1 Generators GN-41 and GN-41-B.- Package each generator Method IIc, as follows: Cushion each generator by wrapping in flexible corrugated paper, securing wrapping with gummed kraft tape, conforming to Specification UU-T-111. Inclose each cushioned generator together with the required amount of desiccant as prescribed in Specification MIL-P-116, within a close-fitting bag fabricated of water-vaporproof barrier material, qualified for class d application conforming to Specification MIL-B-131. Properly seal closure. Place the moisture-vaporproofed generator within a close-fitting suitable style and grade fiberboard box conforming to Specification JAN-P-108. Seal entire box closure with water resistant pressure sensitive tape, conforming to Specification JAN-P-127.

5.2 Packing.-

5.2.1 For domestic shipment.- Generators packaged as specified in 5.1, shall be packed in substantial containers of the type, size, and kind commonly used for the purpose, in such a manner as to insure acceptance at and safe delivery to the designated point by common or other carrier at the lowest rate.

5.2.2 For export shipment.- Place a quantity of generators packaged as specified in 5.1.2.1, together within a wood cleated plywood nailed wood or wire bound wood box conforming to the requirements of JAN-P-105, JAN-P-106, and MIL-B-107 respectively. Fabricate the boxes to fit contents snugly. The approximate gross weight of any exterior shipping container shall not exceed 150 pounds. Close the shipping container as specified in the appendix of the applicable box specification.

5.2.2.1 Shipping containers shall be strapped in conformance with the requirements of the appendix of the applicable container specification only for direct shipment to ports.

5.3 Marking for shipment.- Interior packages and exterior shipping containers shall be marked in accordance with applicable provisions of MIL-STD-129.

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6. NOTES

6.1 Intended use.- Generators GN-41 and GN-41-B are hand operated ringing generators for use with telephone sets. Generator GN-41 is designed for use with telephone sets which do not have steel chassis to dissipate the magnetic field while Generator GN-41-B is designed for use with telephone sets that have steel chassis.

6.2 Ordering data.- Procurement documents should specify the following:

(a) Title, number and date of specification.

(b) Whether domestic or overseas shipment is required.

(c) Samples required (see 3.9).

(d) Marking and shipping of samples.

(e) Place of final inspection (see 4.1).

(f) Acceptable quality level (see 4.1.1 (b), 4.1.2 and Table I).

6.3 Contractual requirements.- It is recommended that the following be included in the contract.

6.3.1 Additional testing.- It should be understood that additional inspection and non-destructive testing of equipment may be performed by the Government inspector when considered necessary to determine compliance with this specification and other applicable documents. The Government inspector, therefore, may withdraw materials, parts, subassemblies or components temporarily from production for such inspection and testing, performed either at a Government laboratory or the contractor's plant.

6.3.2 Waiver of preproduction samples.- When items for which preproduction samples are required are currently in production on a Government order, the contractor may apply to the contracting officer for waiver of preproduction samples on the new order. Any request for waiver should include a statement indicating that (a) the specified samples have been submitted and approved on the current contract, (b) production of the item under the new order will be concurrent with, or follow without interruption, production on the current order, and (c) current production is in strict compliance with specified requirements for the new equipment, or describing any deviations authorized on the current order. Waiver of preproduction samples may be contingent on renegotiation of the contract to reflect the saving in cost to the contractor.

NOTICE: When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other details is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any right or permission to manufacture, use or sell any patented invention that may in any way be related thereto.

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